

BRONTO UI EXPANSION AUDIT

Local audit of every shipped surface, example, package contract, quality gate, and current design-system market signal. Focus: what to add next without breaking the CSS-first, zero-runtime contract.

GENERATED: 2026-06-12 • PROFILE: DECISION REPORT • EVIDENCE STATE: SUPPORTED

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01 / SUMMARY

Bronto UI is already unusually strong for a solo-maintained CSS-first system: the package exports a broad leaf CSS surface, machine-readable tokens/classes, renderer themes, thin framework bindings, packed consumer examples, visual/e2e gates, dead-code checks, and local security scans. This audit's adoption-infrastructure findings have now been implemented locally: a Tailwind v4 CSS theme bridge, Svelte and Vue lifecycle adapters over existing behaviors, a packed Vue example, a packed Tailwind example, and a local Figma Variables handoff built from the existing DTCG token export. Later passes also added durable background-job state and an accessible workbench splitter, so long-running async work and real tool panes now have shared primitives instead of one-off local patterns. The local browser verification paths are now named npm scripts, so the audited evidence is repeatable without translating CI YAML by hand.

READER OUTCOME

ADOPTION BRIDGES, DURABLE ASYNC STATUS,
AND SPLITTER PANES ARE IMPLEMENTED

Basis: local package inventory, roadmap/architecture review, deterministic quality gates, verified 2026 Tailwind/design-system references, the current implementation diff, `npm run test:e2e:nonpixel` browser run, a full `npm run test:examples` packed tarball/example run, cross-browser `ui-job` checks, and cross-browser `ui-splitter` keyboard/pointer/forced-colors checks. Next action: keep the new bridge/adapters/handoff, background-job primitive, and splitter behavior covered by package, example, and browser-smoke gates while resisting a broad component catalog sprint. Recheck: 2026-06-26. Evidence state: supported.

BASIS	45 CSS leaves, 21 delegated behaviors, 634 classes, 9 packed examples, Tailwind/Svelte/Vue exports, Figma Variables artifact, clean local dead-code/AI-slop scans, 245 unit/contract passes, 597 non-pixel Playwright passes, and 9 packed example builds.
NEXT ACTION	Run `npm run test:e2e:nonpixel` and `npm run test:examples` before release-sensitive adoption changes.
RECHECK	2026-06-26

Evidence state: supported · Sources: [S1](#) , [S4](#) , [S5](#)

02 / FINDINGS

RESOLVED FINDING: TAILWIND V4 SUPPORT NOW HAS A CSS BRIDGE

Claim: Bronto had Tailwind layer-order documentation, but no Tailwind v4 `@theme` export, custom variants, or packed Tailwind consumer example; the current tree now ships all three.

Impact: Tailwind v4 made CSS-first theme variables, `@custom-variant`, `@reference`, and `@source` central adoption primitives. Bronto now exposes those adoption affordances without adding a component runtime.

Remediation: the CSS-only `@ponchia/ui/tailwind` export uses `@theme inline`, documents `@reference`/`@source`, and is covered by the packed `examples/tailwind-vite` smoke.

Bronto now ships a Tailwind v4 CSS bridge before adding more components.

SCOPE: `@PONCHIA/UI` PACKAGE EXPORTS, TAILWIND INTEROP DOCS, AND PACKED EXAMPLES AS OF 2026-06-12 AFTER THIS PASS.

EVIDENCE STATE: SUPPORTED.

Basis: local export/doc inventory now includes the bridge; Tailwind's current docs center CSS theme variables, source registration, and reference-only imports.

SOURCES: [S1](#), [S4](#), [S5](#)

RESOLVED FINDING: BEHAVIOR ADAPTERS NOW COVER SVELTE AND VUE

Claim: Bronto already shipped React, Solid, and Qwik lifecycle bindings; the current tree now adds Svelte lifecycle actions, Vue directives/plugin helpers, and a packed Vue example.

Impact: cross-framework consumers can now adopt the delegated behavior layer in the same CSS-first style without Bronto becoming a component library.

Remediation: `@ponchia/ui/svelte` actions and `@ponchia/ui/vue` directives/plugin helpers call the existing delegated behavior initializers and return cleanup; markup, state, and styling stay in consumer land.

Thin Svelte and Vue lifecycle adapters fit Bronto's architecture better than per-framework components or Web Components, and now ship locally.

SCOPE: BEHAVIOR-LAYER ADOPTION SURFACES AND EXAMPLES, NOT NEW COMPONENT MARKUP.

EVIDENCE STATE: SUPPORTED.

Basis: architecture explicitly favors SSR-safe delegated behaviors and optional thin bindings; the local example matrix now includes Vue and the SvelteKit example uses the Svelte adapter.

SOURCES: [S1](#), [S2](#), [S6](#)

RESOLVED FINDING: TOKEN EXPORT NOW HAS A LOCAL FIGMA HANDOFF

Claim: Bronto already emitted DTCG and resolved token artifacts; the current tree now adds a local Figma Variables mapping artifact, docs, and a freshness gate.

Impact: the standards direction is token interoperability. The local Figma Variables export makes the existing source-of-truth story credible without adding bundled Style Dictionary or a design-tool runtime dependency.

Remediation: `tokens/figma.variables.json` is generated from the token source, exported, drift-checked, and documented with permission limits and manual import/export flow.

The token-system improvement is now a local Figma Variables handoff, not a bundled Style Dictionary dependency.

SCOPE: TOKEN EXPORTS, DOCS, AND OFFLINE HANDOFF ARTIFACTS.

EVIDENCE STATE: SUPPORTED.

Basis: Bronto already ships DTCG tokens; DTCG describes cross-tool exchange, and Figma's Variables API supports CI and source-of-truth sync.

SOURCES: [S1](#), [S2](#), [S7](#)

MINOR FINDING: COMPONENT-CATALOG PRESSURE SHOULD STAY DEMAND-GATED

Claim: competing UI kits list much broader component catalogs, but Bronto's local roadmap deliberately centers generated reports, explanation, provenance, and CSS primitives.

Impact: copying shadcn/Radix breadth would dilute the package and conflict with explicit out-of-scope items such as per-framework component packages, Storybook dependency, chart engine, virtualized data-grid, and second accent.

Remediation: keep date/time picker and richer data-table work as recipes until a real app consumer proves demand; do not add a generic catalog sprint.

Bronto should not chase a broad shadcn/Radix-style catalog as the next SOTA move.

SCOPE: ROADMAP PRIORITIZATION AND PUBLIC PACKAGE SURFACE.

EVIDENCE STATE: SUPPORTED.

Basis: external catalogs show the breadth available elsewhere, while Bronto's roadmap explicitly preserves a narrower report/provenance lane.

SOURCES: [S2](#), [S6](#)

RESOLVED FINDING: BACKGROUND JOBS NOW HAVE A PERSISTENT STATE ROW

Claim: current progress/status guidance across comparable systems covers progress indicators and status semantics, but Bronto's state lane still lacked a persistent row for imports, exports, sync runs, and report generation that continue after the initiating interaction is gone.

Impact: without a durable state object, hosts would keep inventing local progress rows or hiding long-running work in toasts, creating inconsistent async UX across report and app surfaces.

Remediation: `css/state.css` now ships `.ui-job`, state modifiers, compact layout, `--job-progress`, forced-colors handling, and reduced-motion-safe progress updates. The class recipe, generated metadata, docs, demo, changelog, roadmap, and cross-browser Playwright checks were updated with explicit host-owned polling, retry, cancellation, and queue semantics.

Bronto now covers persistent background-job status as a CSS-first state primitive instead of leaving async work hidden in transient toasts.

SCOPE: `@PONCHIA/UI/CSS/STATE.CSS`, GENERATED CLASS/TYPE METADATA, STATE DOCS, STATE DEMO, AND CROSS-BROWSER NON-PIXEL TESTS.

EVIDENCE STATE: SUPPORTED.

Basis: `.ui-job` is in source and dist CSS, the class registry exposes `ui.job()`, docs and demo show determinate/progressbar usage, and Chromium/Firefox/WebKit each passed progress-width, ARIA, and forced-colors checks.

SOURCES: [S1](#), [S3](#), [S8](#)

RESOLVED FINDING: WORKBENCH PANES NOW HAVE AN ACCESSIBLE SPLITTER

Claim: Bronto's workbench lane shipped inspector, property, and selection-bar primitives, but still left accessible split panes to each app. Current ARIA references and comparable libraries converge on a focusable separator with keyboard resizing and value semantics.

Impact: without a shared splitter, every real editor/workbench surface would improvise pane layout, pointer drag, keyboard resize, and `aria-valuenow` handling.

Remediation: `css/workbench.css` now ships `.ui-splitter`, pane and handle parts, vertical/horizontal modifiers, and documented sizing knobs. `initSplitter()` wires `data-bronto-splitter` hosts with SSR-safe keyboard and pointer resizing, ARIA value sync, cleanup, and `bronto:splitter:resize`; React, Solid, Qwik, Svelte, and Vue adapters expose the matching lifecycle wrapper.

Bronto now covers accessible workbench split panes without becoming a drag-and-drop or layout-persistence framework.

SCOPE: `@PONCHIA/UI/CSS/WORKBENCH.CSS`, `@PONCHIA/UI/BEHAVIORS`, GENERATED CLASS/TYPE METADATA, WORKBENCH DOCS/DEMO, FRAMEWORK BINDINGS, AND CROSS-BROWSER NON-PIXEL TESTS.

EVIDENCE STATE: SUPPORTED.

Basis: `.ui-splitter` is in source and dist CSS, `initSplitter` exports through the behavior barrel and every binding adapter, docs and demos include separator ARIA, and Chromium/Firefox/WebKit each passed keyboard, pointer, and forced-colors splitter checks.

SOURCES: [S1](#), [S3](#), [S9](#)

MINOR FINDING: LOCAL DETERMINISTIC QUALITY GATES ARE CURRENTLY CLEAN

Claim: native dead-code, lint, unit tests, type check, Semgrep deep, gitleaks, TruffleHog, Trivy, Checkov, clone scanning, and the non-pixel Playwright browser suite all passed locally in this audit; the packed-example browser-smoke matrix also passed through the new local runner.

Impact: the audit did not find confirmed dead code, dependency waste, secret leakage, or AI-slop failures that should block the adoption-roadmap work.

Remediation: keep the existing `check:dead` and declaration-freshness gates in `npm run check`; for new adapters add packed examples, behavior tests, and the named `test:e2e:nonpixel` / `test:examples` local gates rather than relying on static type declarations alone.

The current local quality gate found no confirmed AI-slop or dead-code defects, and the unit, non-pixel Playwright, plus packed-example browser suites passed.

SCOPE: LOCAL DETERMINISTIC SCANS, UNIT/CONTRACT TESTS, REAL-BROWSER NON-PIXEL SUITE, AND PACKED-EXAMPLE SUITE RUN ON 2026-06-12.

EVIDENCE STATE: SUPPORTED.

Basis: the fresh AI-slop follow-up reported 12 helper checks passed, 0 failed; the native `knip` dead-code check passed; `npm test` passed 245 tests; 597 non-pixel Playwright tests passed with 9 intentional browser-feature skips; 9 packed example builds and 7 example browser smokes passed via `npm run test:examples`. Follow-up cleanup added splitter declaration freshness and reduced repeated script server/smoke helpers without introducing new failures.

SOURCE: [S3](#)

MINOR FINDING: DOCUMENTATION DRIFT FOUND DURING THE AUDIT WAS FIXED

Claim: `docs/theming.md` still described the dark `--accent-strong` mix as 84% white, and controlled-modal docs/generated metadata still described focus management as host-owned or absent even though `initModal()` now owns the inert trap, focus-return, and Escape signal. Later in the same audit, `CHANGELOG.md` also missed the unreleased adoption/gate work, `ROADMAP.md` still claimed reconciliation against `0.6.5`, and adapter-contract docs still described parts of the surface as React/Solid/Qwik-only.

Impact: both issues were small but public, because theming and modal behavior are shipped contract surfaces.

Remediation: the theming doc now says 83% black / 80% white; the generated reference, type comments, classes metadata, and usage table now state that `ui.modal({ open: true })` is visual state only and `initModal()` provides controlled-modal focus management while the host still owns visibility, backdrop, and top-layer stacking. The unreleased changelog and roadmap now record the adoption bridges and local verification gates; the adapter docs now include Svelte/Vue where those adapters are part of today's contract, and the Vue adapter has a derived directive-surface regression test.

The confirmed documentation, changelog, and roadmap drifts found in this pass were fixed locally.

SCOPE: `DOCS/THEMING.MD` ACCENT-STRENGTH WORDING; CONTROLLED-MODAL USAGE/REFERENCE, TYPE-COMMENT, AND CLASSES-METADATA WORDING; ADAPTER-CONTRACT DOCS; `CHANGELOG.MD` UNRELEASED NOTES; AND `ROADMAP.MD` RECONCILIATION STATUS.

EVIDENCE STATE: SUPPORTED.

Basis: local generated token artifacts, behavior docs, generated metadata, adapter docs, changelog, and roadmap were compared; the affected docs, generator sources, and derived adapter tests were patched, then regenerated where applicable.

SOURCE: S1

03 / EVIDENCE LEDGER

Claim, evidence, source, and caveat ledger			
CLAIM	EVIDENCE	SOURCE	CAVEAT
C1	Exports/docs now include a Tailwind v4 bridge; Tailwind v4 docs center CSS theme variables.	S1 , S4	Implementation should keep the bridge CSS-only and optional.

CLAIM	EVIDENCE	SOURCE	CAVEAT
C2	React/Solid/Qwik/Svelte/Vue adapters exist; Vue has derived directive-surface coverage; Vue and SvelteKit examples smoke through packed consumers.	S1 , S2	Adapters must not become component packages.
C3	DTCG and Figma Variables exports exist locally; token standards and Figma APIs support cross-tool handoff.	S1 , S7	Figma API write access requires Enterprise permissions; ship local artifacts and docs first.
C4	Roadmap says the report/provenance lane is strategic and rejects several broad-kit dependencies.	S2 , S6	Real app demand can still promote recipes into primitives later.
C5	The state CSS, generated classes/types, docs, demo, and Playwright spec now cover durable background-job rows.	S1 , S3 , S8	The host still owns polling, retries, queue position, cancellation, and partial failure semantics.
C6	Workbench CSS, behavior exports, generated metadata, docs, demo, framework bindings, and browser specs now cover accessible split panes.	S1 , S3 , S9	The host still owns pane content, persistence, collapse policy, saved layout state, and any drag/drop semantics.
C7	Dead-code and deep deterministic AI-slop helper reports show passes and no failures; `npm test`, `npm run test:e2e:nonpixel`, and `npm run test:examples` passed.	S3	Mutation and external reviewer were intentionally not run in this local-only pass; CodeQL was triaged separately in the AI-slop pass.

CLAIM	EVIDENCE	SOURCE	CAVEAT
C8	The theming doc, controlled-modal docs/generated metadata, adapter docs, changelog, and roadmap now match the audited implementation.	S1	Final `npm run check`, focused generation gates, `npm test`, `npm run test:e2e:nonpixel`, and `npm run test:examples` passed after the edits.

04 / ACTIONS

Implement Tailwind v4 CSS bridge and packed example.

STATUS: COMPLETE

OWNER: MAINTAINER

DUE/RECHECK: 2026-06-19

PRIORITY: HIGH

Success criteria: `@ponchia/ui/tailwind` exports a CSS-first `@theme inline` bridge; docs cover layer order, variants, `@reference`, and `@source`; `examples/tailwind-vite` builds from `npm pack` and browser-smokes in CI.

SOURCE: C1

Add Svelte and Vue lifecycle adapters over behaviors.

STATUS: COMPLETE

OWNER: MAINTAINER

DUE/RECHECK: 2026-07-03

PRIORITY: HIGH

Success criteria: `@ponchia/ui/svelte` and `@ponchia/ui/vue` expose lifecycle glue only; the SvelteKit example stops manual behavior wiring; a Vue Vite example builds from the tarball and has a browser smoke path.

SOURCE: C2

Create local Figma Variables handoff docs and artifact.

STATUS: COMPLETE

OWNER: MAINTAINER

DUE/RECHECK: 2026-07-10

PRIORITY: MEDIUM

Success criteria: generated Figma Variables mapping derives from `tokens/tokens.dtcg.json`, is drift-checked, and docs explain permission limits and manual import/export flow.

SOURCE: C3

Add persistent background-job state primitive.

STATUS: COMPLETE

OWNER: MAINTAINER

DUE/RECHECK: 2026-06-26

PRIORITY: MEDIUM

Success criteria: `.ui-job` provides queued/running/blocked/failed/complete state rows, determinate progress via `--job-progress`, forced-colors visibility, documentation, demo markup, generated class metadata, and browser tests.

SOURCE: C5

Add accessible workbench splitter primitive and behavior.

STATUS: COMPLETE

OWNER: MAINTAINER

DUE/RECHECK: 2026-06-26

PRIORITY: MEDIUM

Success criteria: `.ui-splitter` provides vertical/horizontal split panes, documented separator ARIA, keyboard and pointer resizing through `initSplitter`, framework binding parity, generated class/contract metadata, demo coverage, and Chromium/Firefox/WebKit non-pixel tests.

SOURCE: C6

Keep broad components recipe-only until demanded.

STATUS: OPEN

OWNER: MAINTAINER

DUE/RECHECK: 2026-07-10

PRIORITY: MEDIUM

Success criteria: no Storybook dependency, no per-framework component package, no chart engine, no virtualized data-grid, and no second accent are introduced while completing the adoption bridge work.

SOURCE: C4

SOURCES

● S1 verified ● S2 verified ● S3 verified ● S4-S9 verified ● 8 claims mapped

LOCAL REPOSITORY/PACKAGE INVENTORY

Verified: `package.json`, exported paths, `docs/interop/tailwind.md`, examples, framework bindings, token artifacts, size report, the local theming-doc patch, controlled-modal documentation/generator patches, adapter-doc cleanup, the derived Vue directive-surface test, background-job state primitive/docs/demo/tests, workbench splitter primitive/docs/demo/tests, splitter declaration freshness, shared static-server helper cleanup, smoke assertion helper cleanup, changelog/roadmap reconciliation, and local browser-gate script docs.

2026-06-12

Observed 45 CSS leaf files, 21 delegated behavior exports, 634 generated classes, 62 CSS custom properties, 39 behavior attributes, 9 consumer examples, Tailwind/Svelte/Vue adapter exports, and a Figma Variables artifact. The state lane now includes `.ui-job` with durable async status modifiers; the workbench lane now includes `.ui-splitter` with an ARIA separator behavior.

LOCAL ARCHITECTURE AND ROADMAP DOCTRINE

Verified: `CONTRIBUTING.md`, `docs/architecture.md`, and `ROADMAP.md` read locally.

2026-06-12

The project is CSS-first, framework-agnostic, zero-runtime by default, and explicitly avoids Storybook dependency, per-framework component packages, bundled Style Dictionary, chart engine, virtualized data-grid, and second accent.

LOCAL DETERMINISTIC QUALITY SCANS

Verified: repo-local reports at `code-quality-audits/2026-06-12-expansion-research/dead-code/report.md`, `code-quality-audits/2026-06-12-expansion-research/ai-slop/report.md`, latest follow-up reports under `code-quality-audits/2026-06-12-ai-slop-followup-3/` and `code-quality-audits/2026-06-12-dead-code-followup-3/`, plus the named unit, non-pixel Playwright, and packed-example runs on 2026-06-12.

2026-06-12

Native `knip` dead-code passed. Deep deterministic AI-slop pass reported 12 passes, 0 failures, and 4 skipped checks; skips were no typecheck script, no raw knip after native coverage, no dependency-cruiser config, and no OpenAPI/Spectral path. `npm run check` passed. `npm test` passed 245 tests. Non-pixel Playwright coverage passed 597 tests with 9 intentional skips; the packed tarball example gate built 9 examples and browser-smoked 7. Source-scoped Trivy was clean; jscpd script clones were reduced from 13 to 11 after extracting shared static-server and smoke assertion helpers.

TAILWIND CSS V4 OFFICIAL ANNOUNCEMENT

Verified: official Tailwind blog, retrieved 2026-06-12. Tailwind CSS v4.0

2026-06-12

Tailwind v4 emphasizes modern CSS, first-party Vite integration, automatic content detection, built-in imports, CSS-first configuration, and CSS theme variables.

TAILWIND CSS THEME/DIRECTIVE/SOURCE DOCS

Verified: official Tailwind docs for theme variables, directives, source detection, and upgrade requirements, retrieved 2026-06-12. Theme · Directives · Source detection · Upgrade guide

2026-06-12

The docs map `@theme` namespaces to utilities, recommend `@theme inline` for variables that reference other variables, document `@custom-variant` and `@reference`, and define modern browser requirements.

COMPARABLE DESIGN-SYSTEM ADOPTION SURFACES

Verified: public docs for Radix Themes, shadcn/ui, Ark UI, Shoelace, and React Aria, retrieved 2026-06-12. Radix Themes · shadcn/ui · Ark UI · Shoelace · React Aria

2026-06-12

The market pattern is clear: token variables, Tailwind-aware states, and cross-framework adoption paths matter. Bronto should copy those adoption affordances without copying the large component-catalog strategy.

TOKEN EXCHANGE AND FIGMA VARIABLES STANDARDS

Verified: DTCG 2025.10 format report and Figma Variables REST API docs, retrieved 2026-06-12. DTCG format · Figma Variables API

2026-06-12

DTCG is a stable exchange format for design-token data. Figma Variables API documentation says variables can integrate with CI and sync a design-system source of truth, subject to enterprise seat and token-scope requirements.

PROGRESS AND STATUS PATTERN REFERENCES

Verified: Atlassian, Carbon, Material Design 3, Material Web, and background-work UX references retrieved 2026-06-12. [Atlassian progress](#) · [Carbon progress](#) · [Carbon status](#) · [Material progress](#) · [Material Web progress](#)

2026-06-12

The common pattern is explicit progress/status feedback with accessible semantics. Bronto's local implementation adapts that into a persistent CSS state row while keeping the async state machine in the host.

ACCESSIBLE SPLITTER AND RESIZE-PANE REFERENCES

Verified: WAI-ARIA APG Window Splitter, MDN separator role, React Resizable Panels, Salt Splitter accessibility, Radix Vue Splitter, and shadcn Resizable docs retrieved 2026-06-12. [WAI-ARIA APG](#) · [MDN separator](#) · [React Resizable Panels](#) · [Salt Splitter](#)

2026-06-12

The shared shape is a focusable separator that exposes orientation, value bounds and current value, supports keyboard resizing, and leaves persistence/content state to the host application.